

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Summative Assessment 1

Summative Assessment (SA-1)

Class: Class 8

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Functions, Data Structures, SQL, HTML/CSS

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

- a. Which keyword defines a function in Python?
 - b. func
 - c. def
 - d. function
 - e. define
- b. What is the output of `len([1, [2, 3], 4])`?
 - c. 4
 - d. 3
 - e. Error
 - f. 5
- c. Which SQL command deletes all rows but keeps the table structure?
 - d. DELETE
 - e. DROP
 - f. TRUNCATE
 - g. REMOVE
- d. In HTML, which tag creates a table row?
 - e. `<tr>`
 - f. `<td>`
 - g. `<th>`
 - h. `<row>`
- e. Which CSS selector has the highest specificity?
 - f. Class selector `.box`
 - g. Element selector `p`
 - h. ID selector `#main`
 - i. Universal selector `*`
- f. What will be the output?

```
def f(x, y=[]):
    y.append(x)
    return y
```

```
print(f(1))
print(f(2))
```

- a. [1] then [2]
- b. [1] then [1, 2]
- c. [1] then [1]
- d. Error
- g. Which method returns a new sorted list without modifying the original?
- h. `sort()`
- i. `sorted()`
- j. `reverse()`
- k. `reversed()`
- h. Which SQL query displays only unique records?
- i. `SELECT * FROM table;`
- j. `SELECT UNIQUE * FROM table;`
- k. `SELECT DISTINCT * FROM table;`
- l. `SELECT DIFFERENT * FROM table;`
- i. Which CSS property adds space between the element border and its content?
- j. `margin`
- k. `padding`
- l. `border-spacing`
- m. `gap`
- j. Which of the following is a mutable data structure in Python?
- k. Tuple
- l. String
- m. List
- n. Integer

Part B: Short Answer (8 x 2 = 16 marks)

- a. Write a Python function `reverse_string(s)` that returns the reversed version of string `s`. Include proper handling for empty strings.
- b. Write the output of the following and explain line by line:

```
def factorial(n):
    if n <= 1:
        return 1
    return n * factorial(n - 1)
```

```
print(factorial(0))
print(factorial(5))
print(factorial(3))
```

- c. Differentiate between `list.append()` and `list.extend()` with examples. What happens when you append a list to another list?
- d. Write a Python program to count the frequency of each word in the sentence "the cat sat on the mat the cat" using a dictionary. Display the result.
- e. Write SQL queries for a table `Employees(emp_id, name, department, salary, joining_date)`:
- f. Find employees with salary between 30000 and 50000
- g. Count employees per department
- h. Find the employee with the maximum salary
- i. Delete employees who joined before 2020
- f. Differentiate between `DELETE` and `TRUNCATE` with three points each.
- g. Write the HTML code for a form with: name, email, date of birth (date input), gender (radio buttons), and a submit button. Include proper labels.
- h. Write CSS rules to:
 - i. Create a responsive navigation bar with flexbox
 - j. Style links with hover underline effect
 - k. Set a media query for screens below 768px width

Part C: Long Answer (4 x 3 = 12 marks)

- a. Write a complete Python program that uses functions and lists to manage a shopping cart. The program should:
 - Allow adding items with name, price, and quantity
 - Remove items by name
 - Calculate the total bill with 18% GST
 - Display a formatted receipt with itemised details
- b. Write SQL queries for a School database with tables: `Students(roll, name, class, section)`, `Subjects(sub_id, sub_name, max_marks)`, `Marks(roll, sub_id, marks_obtained)`:
- c. List all students who scored above 80 in any subject
- d. Find the percentage of each student (total marks obtained / total max marks)
- e. Find the subject with the highest average score
- f. List students ranked by total marks descending
- c. Create a complete HTML page for a **Student Report Card** that:
 - Has a header with school name and logo placeholder
 - Includes a styled table with columns: Subject, Marks Obtained, Grade
 - Shows summary statistics (Total, Percentage, Class Rank)
 - Uses CSS for professional formatting with borders, colours, and spacing
 - Has a print-friendly layout

d. Write a Python program using dictionaries and functions to create a simple **Quiz Application** that:

- Stores at least 5 questions with 4 options and correct answers
- Presents questions to the user one by one
- Accepts user input and validates it
- Keeps track of score and displays final results with percentage and grade

Part D: Application Based (2 x 1 = 2 marks)

a. A student writes the following code and gets unexpected output:

```
def add_item(item, cart=[]):  
    cart.append(item)  
    return cart  
  
cart1 = add_item("apple")  
cart2 = add_item("banana")  
print(cart1)  
print(cart2)
```

Identify the bug, explain why it occurs, and write the corrected version.

b. Read the following SQL query and answer:

```
SELECT department, COUNT(*) as emp_count, AVG(salary) as avg_salary  
FROM Employees  
WHERE joining_date > '2020-01-01'  
GROUP BY department  
HAVING AVG(salary) > 40000  
ORDER BY avg_salary DESC;
```

Explain what this query does step by step and what each clause's role is.